R16

Code No:131AH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD
B.Tech I Year I Semester Examinations, December - 2018
ENGINEERING PHYSICS-I
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Part- A

(25 Marks)

1.a)	Define interference.	[2]
b)	What is diffraction	[3]
c)	Explain the phenomenon of polarization of light.	[2]
d)	Explain spontaneous emission.	[3]
e)	Write any two applications of optical fiber.	[2]
f)	What is optical fiber? Explain basic principle of optical fiber	[3]
g)	Define Space lattice.	[2]
h)	Distinguish between unit cell and primitive cell.	[3]
i)	What are the conditions to occur diffraction due to X-rays?	[2]
j)	Define Burger's vector.	[3]

Part-B

(50 Marks)

2.a)	Discuss interference in thin film due to transmitted light.	
b)	Write a note on Newton's ring experiment.	
c)	What is Coherence?	[4+4+2]

OR

3.a)	Write a note on division of amplitude.	
b)	Distinction between Fresnel and Fraunhofer diffraction.	
c)	Discuss diffraction due to N- silts.	[3+3+4]

4.a)	Explain Malus law.	
b)	What is double refraction?	
c)	Explain the terms population and population inversion.	[3+3+4]

OR

5.a)	Explain Quarter wave plate.	
b)	With neat diagram explain working principle of Ruby laser system.	
c)	Write any two applications of lasers.	[3+5+2]

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6.a) With neat diagram explain construction of optical fiber.
b) Derive an expression for Acceptance angle.
c) Derive an equation for Numerical aperture. [4+3+3]

OR

7.a) Explain how Step index optical fibers are classified.
b) Write a note on graded index optical fiber.
c) What are the medical applications of optical fiber? [4+4+2]

8.a) With neat diagram explain types of crystal systems.
b) Define crystal plane.
c) Discuss construction of Diamond structure. [4+2+4]

OR

9.a) What are the Bravais lattices?
b) What are the Miller indices?
c) Derive an expression for inter planner spacing of orthogonal crystal system. [4+3+3]

10.a) Derive an equation for Bragg's law.
b) Discuss powder method.
c) Explain vacancy defect. [4+4+2]

OR

11.a) Discuss Laue method.
b) What is Frenkel defect?
c) Write a note on Schottky defect. [4+3+3]

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